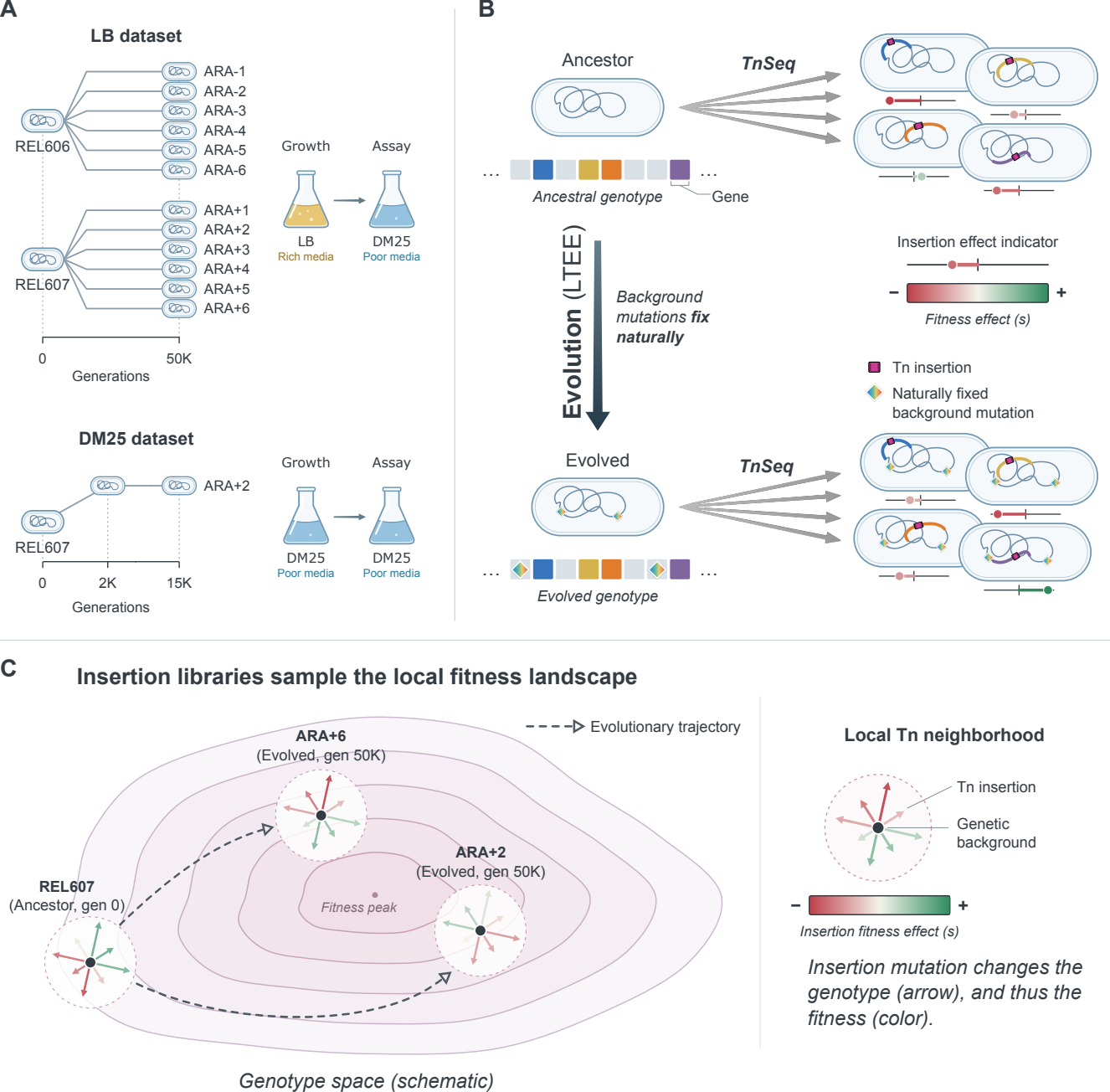

A
LB dataset
REL606
ARA-1
ARA-2
ARA-3
ARA-4
ARA-5
ARA-6
REL607
ARA+1
ARA+2
ARA+3
ARA+4
ARA+5
ARA+6
0
50K
Generations
Growth
Assay
LB
Rich media
DM25
Poor media
DM25 dataset
REL607
ARA+2
0
2K
15K
Generations
Growth
Assay
DM25
Poor media
DM25
Poor media
B
Ancestor
TnSeq
Ancestral genotype
Gene
Evolution (LTEE)
Background mutations fix naturally
Insertion effect indicator
Fitness effect (s)
Tn insertion
Naturally fixed background mutation
Evolved
TnSeq
Evolved genotype
C
Insertion libraries sample the local fitness landscape
Evolutionary trajectory
ARA+6
(Evolved, gen 50K)
ARA+2
(Evolved, gen 50K)
REL607
(Ancestor, gen 0)
Fitness peak
Genotype space (schematic)
Local Tn neighborhood
Tn insertion
Genetic background
Insertion fitness effect (s)
Insertion mutation changes the genotype (arrow), and thus the fitness (color).